

# Supplementary for Investment Theory

## Abstract

This lecture note introduces key concepts related to the time value of money. While these concepts are typically covered in financial management textbooks, they are also relevant to investment decisions in macroeconomics. Therefore, I will provide simple examples to help you understand these concepts more easily.

## Future Value, FV

**Definition 0.1.** Let  $C$  be the cash flow in today,  $FV_n$  be the future value of the cash flow,  $r$  be the constant interest rate,  $n$  be the periods, then

$$FV_n = C \times (1 + r) \times (1 + r) \times \cdots \times (1 + r) = C \times (1 + r)^n$$

Suppose we want to know how much the \$1,000 is worth in two years' time. If the interest rate for year 2 is also 10%, then we convert as we just did:

$$1,000 \times (1 + 0.1)^2 = 1,210$$

Let's represent this calculation on a timeline as follows:

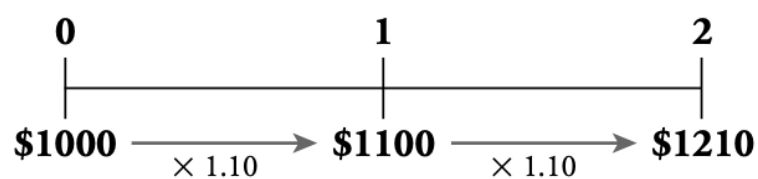


Figure 1: timeline of future value

## Present Value, PV

**Definition 0.2.** Let  $C$  be a fixed cash flow in each period,  $PV_n$  be the present value of the cash flow,  $r$  be the constant interest rate,  $n$  be the periods, then

$$PV_n = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots + \frac{C}{(1+r)^n} = \sum_{i=1}^n \frac{C}{(1+r)^i}$$

Suppose that you anticipate receiving the \$1,000 two years from today rather than in one year. If the interest rate for both years is 10%, then we convert as we just did:

$$\frac{1,000}{(1+0.1)^2} = 826.45$$

Let's represent this calculation on a timeline as follows:

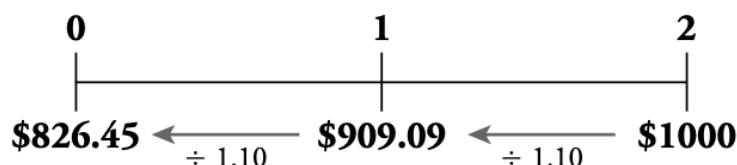


Figure 2: timeline of present value

**Example 1.** You have just graduated and need money to buy a new car. Your rich Uncle Henry will lend you the money so long as you agree to pay him back within four years, and you offer to pay him the rate of interest that he would otherwise get by putting his money in a savings account. Based on your earnings and living expenses, you think you will be able to pay him \$5,000 in one year, and then \$8,000 each year for the next three years. If Uncle Henry would otherwise earn 6% per year on his savings, how much can you borrow from him?

The cash flows you can promise Uncle Henry are as follows:

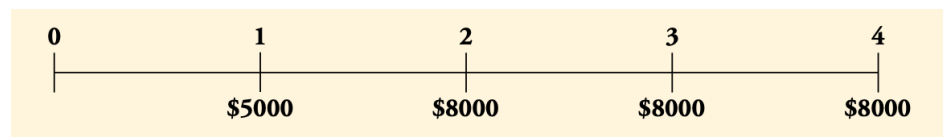


Figure 3: Cashflow

How much money should Uncle Henry be willing to give you today in return for your promise of these payments? He should be willing to give you an amount that is equivalent to these payments in present value terms. This is the amount

of money that it would take him to produce these same cash flows, which we calculate as follows:

$$\begin{aligned} PV &= \frac{5,000}{(1+0.06)} + \frac{8,000}{(1+0.06)^2} + \frac{8,000}{(1+0.06)^3} + \frac{8,000}{(1+0.06)^4} \\ &= 4,716.98 + 7,119.97 + 6,716.95 + 6,336.75 \\ &= 24,890.65 \end{aligned}$$

Thus, Uncle Henry should be willing to lend you \$24,890.65 in exchange for your promised payments. This amount is less than the total you will pay him (\$5,000 + \$8000 + \$8000 + \$8000 = \$29,000) due to the time value of money.

Let's verify our answer. If your uncle kept his \$24,890.65 in the bank today earning 6% interest, in 4 years he would have

$$FV = 24,890.65 \times (1.06)^4 = 31,423.87$$

Now suppose that Uncle Henry gives you the money, and then deposits your payments to him in the bank each year. How much will he have four years from now? We need to compute the future value of the annual deposits. One way to do so is to compute the bank balance each year:

$$FV = 5,000 \times (1+0.06)^3 + 8,000 \times (1.06)^2 + 8,000 \times (1.06) + 8,000 = 31,423.88$$

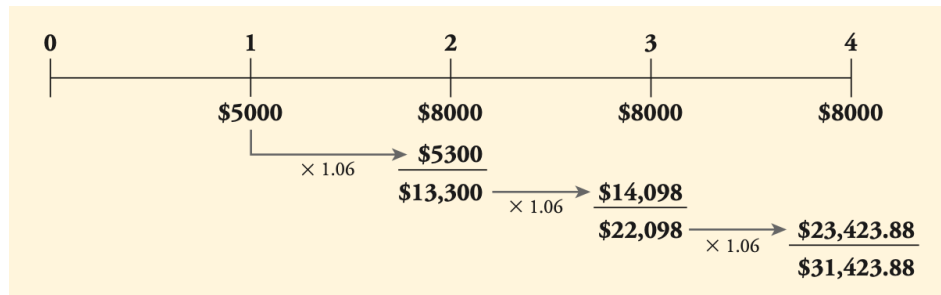


Figure 4: Cashflow of future value



## Net Present Value, NPV

**Definition 0.3.** *Net present value of a project or investment is the difference between the present value of its benefits and the present value of its costs:*

$$NPV = PV(\text{Benefits}) - PV(\text{Costs})$$

If we use positive cash flows to represent benefits and negative cash flows to represent costs, and calculate the present value of multiple cash flows as the sum of present values for individual cash flows, we can also write this definition as

$$NPV = PV(\text{All project cash flows})$$

That is, the NPV is the total of the present values of all project cash flows.

**Example 2.** Suppose your firm is offered the following investment opportunity: In exchange for \$500 today, you will receive \$550 in one year with certainty. If the risk-free interest rate is 8% per year then

$$NPV = -500 + \frac{550}{(1 + 0.08)} = 9.26$$

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Because NPV is expressed in terms of cash today, it simplifies decision making. As long as we have correctly captured all of the costs and benefits of the project, decisions with a positive NPV will increase the wealth of the firm and its investors. We capture this logic in the **NPV Decision Rule**.

**Definition 0.4.** *When making an investment decision, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.*

**Accepting or rejecting a Project.** A common investment decision is whether to accept or reject a project. Because rejecting the project generally has  $NPV = 0$  (there are no new costs or benefits from not doing the project), the NPV decision rule implies that we should

- Accept those projects with positive NPV because accepting them is equivalent to receiving their NPV in cash today.
- Reject those projects with negative NPV; accepting them would reduce the wealth of investors, whereas not doing them has no cost ( $NPV = 0$ ).

If the NPV is exactly zero, you will neither gain nor lose by accepting the project rather than rejecting it. It is not a bad project because it does not reduce firm value, but it does not increase value either.

## Tobin's q theory

**Definition 0.5.**

$$q = \frac{\text{the market value of capital}}{\text{the replacement cost of capital}}$$

**Whenever  $q$  is greater than 1, a firm should add physical capital** because for each dollar's worth of new machinery, the firm can sell stock for  $q$  dollars and pocket a profit  $q - 1$ . This implies a flood of investment whenever  $q > 1$ . **Whenever  $q$  is lower than 1, a firm should decrease physical capital.**